

Astro210 HW 13

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13.1

a)

Given ...

$$E(t) = \frac{4\sigma_{SB}}{c} (T^4 V), \text{quad} P(t) = \frac{4\sigma_{SB}}{3c} T^4$$

We find ...

$$\frac{dE}{dt} \frac{4\sigma_{SB}}{c} (T^4 V) = \frac{4\sigma_{SB}}{c} \left(4T^3 V \frac{dT}{dt} + T^4 \frac{dV}{dt} \right)$$

$$\boxed{\frac{4\sigma_{SB}}{c} \left(4T^3 V \frac{dT}{dt} + T^4 \frac{dV}{dt} \right) = -\frac{4\sigma_{SB}}{3c} T^4 \frac{dV}{dt}}$$

b)

Given ...

$$\frac{4\sigma_{SB}}{c} \left(4T^3 V \frac{dT}{dt} + T^4 \frac{dV}{dt} \right) = -\frac{4\sigma_{SB}}{3c} T^4 \frac{dV}{dt}$$

We find ...

$$\cancel{\frac{4\sigma_{SB}}{c}} \left(4T^3 V \frac{dT}{dt} + T^4 \frac{dV}{dt} \right) = -\cancel{\frac{4\sigma_{SB}}{c}} T^4 \frac{1}{3} \frac{dV}{dt}$$

$$\left(4 \frac{T^3}{T^4} V \frac{dT}{dt} + T^4 \frac{dV}{dt} \right) = -T^4 \frac{1}{3} \frac{dV}{dt}$$

$$4 \frac{1}{T} V \frac{dT}{dt} = -\frac{1}{3} \frac{dV}{dt} - \frac{dV}{dt}$$

$$4 \frac{1}{T} V \frac{dT}{dt} = -\frac{4}{3} \frac{dV}{dt}$$

$$\frac{1}{T} V \frac{dT}{dt} = -\frac{1}{3} \frac{dV}{dt}$$

$$\boxed{\frac{1}{T} \frac{dT}{dt} = -\frac{1}{3V} \frac{dV}{dt}}$$

c)

Given ...

$$\frac{1}{T} \frac{dT}{dt} = -\frac{1}{3V} \frac{dV}{dt}, \quad V(t) \propto a(t)^3$$

We find ...

$$\frac{dV}{dt} = 3a^2 \frac{da}{dt}$$

$$\frac{1}{T} \frac{dT}{dt} = -\frac{1}{a} \frac{da}{dt}$$

$$\boxed{\frac{1}{T} \frac{dT}{dt} = -\frac{1}{a} \frac{da}{dt}}$$

d)

Given ...

$$\frac{1}{T} \frac{dT}{dt} = -\frac{1}{a} \frac{da}{dt}$$

We find ...

$$\frac{1}{T} = \ln(T) dT, \quad \frac{1}{a} = \ln(a) da$$

$$\boxed{\frac{d}{dt} \ln(T) = \ln(a) \frac{d}{dt}}$$

e)

Given ...

$$\lambda = \frac{2900 \mu\text{m}}{T}$$

$$T = \frac{2900 \mu\text{m}}{\lambda}$$

We find ...

$$\frac{d}{dt} \ln \left(\frac{2900 \mu\text{m}}{\lambda} \right) = \ln(a) \frac{d}{dt}$$

$$\frac{d}{dt} \ln(2900 \mu\text{m}) - \ln(\lambda) = \ln(a) \frac{d}{dt}$$

$$\boxed{\ln(\lambda) \propto \ln(a)}$$

f)

Given ...

$$\begin{aligned}\frac{1}{2} \left(\frac{dr}{dt} \right)^3 &= \frac{GM}{r(t)} + k \\ \left(\frac{\dot{a}}{a} \right)^2 &= \frac{8\pi G r_0^2}{3} P(t) + \frac{2\kappa}{r_0^2} \frac{1}{a(t)^2} \\ M - \frac{4\pi}{3} p(t) r(t)^3 \\ \frac{4G\pi r(t)^2 p(t)}{3} + \kappa &= 1/2 \frac{dr^3}{dt} \\ r(t) &= a(t) r_0 \\ \frac{1}{2} r_0^2 \dot{a}^2 &= \frac{4}{3} G\pi (a(t) r_0)^2 + \kappa = \frac{1}{2} \left(\frac{dr}{dt} \right)^2 \\ \boxed{\left(\frac{\dot{a}}{a} \right)^2} &= \frac{8\pi}{3} p(t) + \frac{2k}{r_0^2} \frac{1}{a(t)^2}\end{aligned}$$

g)

Given ...

$$\begin{aligned}\left(\frac{\dot{a}}{a} \right)^2 &= \frac{8\pi}{3} p(t) + \frac{2k}{r_0^2} \frac{1}{a(t)^2} \\ \left(\frac{\dot{a}}{a} \right)^2 &= H(t)^2 \\ k &= 0 \\ H(t)^2 &= \frac{8\pi G}{3c^2} v_c \\ \frac{3c^2 H(t)^2}{8G\pi} &= v_c\end{aligned}$$

h)

i)

13.2 Precision cosmological measurements

a)

Given ...

$$H_{0_1} = 67 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad H_{0_2} = 74 \text{ km s}^{-1} \text{ Mpc}^{-1}$$
$$V_{rec} = d * H_0$$

We find ...

$$V_{rec_1} = 7.504 \times 10^6 \text{ m/s}, \quad V_{rec_2} = 8.288 \times 10^6 \text{ m/s}$$

b)

Given ...

$$c = \lambda \nu$$
$$\lambda = \frac{c}{\nu}$$
$$\nu = 22.23508 \text{ GHz}$$

We find ...

$$\lambda = 1.348 \times 10^{-2} \text{ m}$$

c)

Given ...

$$V_{rec_1} = 7.504 \times 10^6 \text{ m/s}, \quad V_{rec_2} = 8.288 \times 10^6 \text{ m/s}$$
$$\lambda_{obsv} = \lambda_{rest}(1 + z)$$
$$z = \frac{v}{c}$$

We find ...

$$z_1 = 2.503 \times 10^{-2}, \quad z_2 = 2.765 \times 10^{-2}$$
$$\nu_1 = 2.279 \times 10^{10} \text{ 1/s}, \quad \nu_2 = 2.285 \times 10^{10} \text{ 1/s}$$
$$\lambda_1 = 1.315 \times 10^{-2} \text{ m}, \quad \lambda_2 = 1.312 \times 10^{-2} \text{ m}$$
$$\Delta\lambda = 3.347 \times 10^{-5} \text{ m}$$

13.3 Matter-dark energy equality

a)

Given ...

$$\Omega_{\lambda_0} = (1 + z_{eq})^3 \Omega_{m_0}$$

We find ...

$$\frac{\Omega_{\lambda_0}}{\Omega_{m_0}} = (1 + z_{eq})^3$$
$$\left(\frac{\Omega_{\lambda_0}}{\Omega_{m_0}} \right)^{1/3} - 1 = z_{eq}$$
$$z_{eq} \approx 3.931 \times 10^{-1}$$

b)

Given ...

$$\ddot{a} = H_0 \left[\frac{\Omega_{r,0}}{a^2} + \frac{\Omega_{m,0}}{a} + \Omega_{\lambda,0} a^2 \right]^{1/2}$$
$$1 + z = \frac{1}{a} \Rightarrow a = \frac{1}{1 + z}$$

We find ...

$$\boxed{\ddot{a} = H_0(0.86734)}$$

c)

Given ...

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_{r,0}}{a(t)^4} + \frac{\Omega_{m,0}}{a(t)^3} + \Omega_{\Lambda}$$

We find ...

$$H(t) = H_0^2 \left[\frac{\Omega_{m,0}}{a(t)^3} + \Omega_{\Lambda} \right]^{1/2}$$
$$\boxed{H(t) = (H_0)^2 1.2083}$$

13.4 Curvature

Given ...

$$\alpha + \beta + \gamma = \pi + \frac{\kappa A}{r_{c,0}^2}, \quad \text{By spherical} \Rightarrow \kappa = 1$$
$$\alpha + \beta + \gamma = \theta_{tot}$$
$$A = \frac{\sqrt{3}}{4} \ell^2, \quad \ell = 1$$

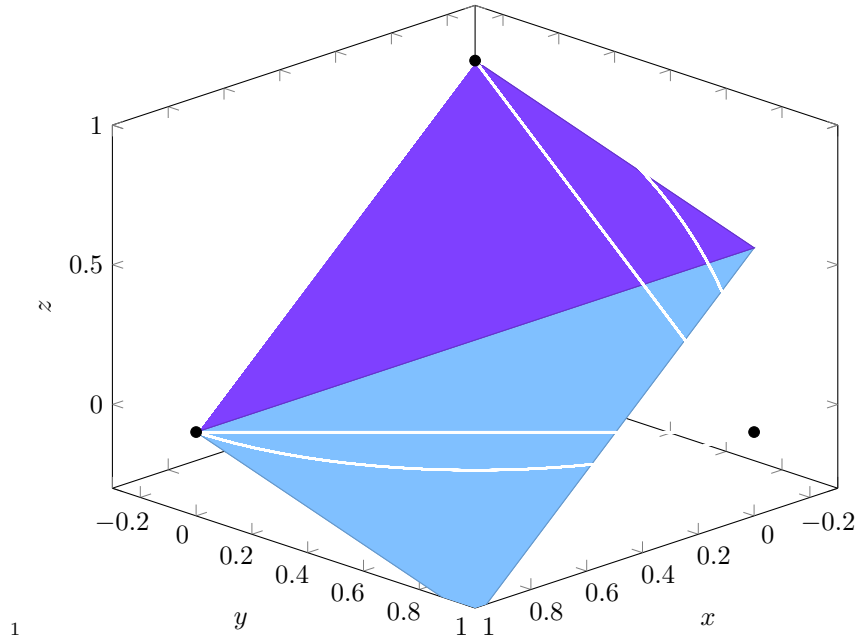
We find ...

$$\theta_{tot} = \pi + \frac{A}{r^2}$$

$$\boxed{\theta_{tot} = 3.141\,592\,664 \text{ rad}}$$

$$\boxed{\theta_{tot} - \pi = 1.066\,806 \times 10^{-8} \text{ rad}}$$

Spherical triangle on a unit sphere



13.5 Proper distances

Given ...

$$r = \int_{t_{eq}}^{t_0} \frac{dt}{a(t)}$$

$$a(t) = \frac{t}{t_0}$$

We find ...

$$r = ct_0 \int_{t_{eq}}^{t_0} \frac{dt}{t}$$

$$r = ct_0 \ln(t) \Big|_{t_{eq}}^{t_0}$$

$$r = ct_0 (\ln(t_0) - \ln(t_{eq}))$$

$$r = ct_0 \left[\ln \left(\frac{t_0}{t_{eq}} \right) \right]$$

$$r = ct_0 \left[\ln \left(\frac{1}{a_{eq}} \right) \right]$$

$$r = ct_0 (\ln(1+z))$$

$$r \approx 4.329 \times 10^{25} \text{ m} \approx 1.403 \times 10^3 \text{ Mpc}$$

¹spent an hour procrastinating the actual homework making this